

# TrES-5b

R-0980

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_TRES-5\_230809 · created 2026-07-29T20:27:46+00:00

BENCHMARK: ACCURATE — fit consistent with published values

## Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2460165.6865 +/- 0.0028 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.17 +/- 0.017                  |
| Transit depth (Rp/Rs)^2                           | 2.89 +/- 0.57 [%]               |
| Orbital Inclination (inc)                         | 86.59 +/- 1.52                  |
| Airmass coefficient 1 (a1)                        | 0.985 +/- 0.022                 |
| Airmass coefficient 2 (a2)                        | 0.008 +/- 0.061                 |
| Scatter in the residuals of the lightcurve fit is | 2.17 %                          |
| Best Comparison Star                              | #1 - [166, 479]                 |
| Optimal Aperture                                  | 3.49                            |
| Optimal Annulus                                   | 8.04                            |
| Transit Duration (day)                            | 0.0799 +/- 0.0038               |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                        |
|---------------------------|----------------------------------------|
| Rp/R■ fit vs published    | 0.17 ± 0.017 vs 0.142 (deviation 1.4σ) |
| Duration fit vs published | 0.0799 d vs 0.0758 d (deviation 0.91σ) |
| Mid-time O–C              | -6.3 min                               |
| Depth SNR                 | 8.5                                    |
| Residual scatter          | 2.17 %                                 |

## Light curve

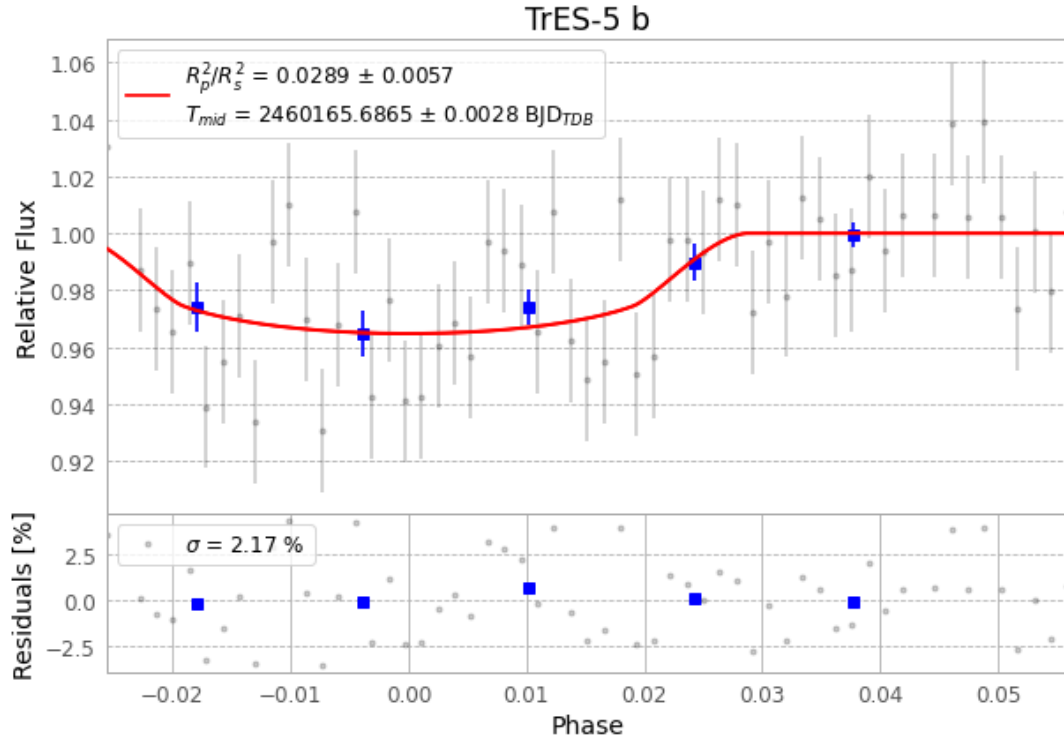


Figure 1 — FinalLightCurve\_TrES-5 b\_2023-08-08.png (SHA-256 875752e1cf5e92b4...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [255, 233], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 d317a86accc78ed1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ead60b9

## Data provenance

58 FITS frames (38 MB), TRES-5230809033018.FITS → TRES-5230809062415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-230809.log (e4a7d6457796...), FinalLightCurve\_TrES-5 b\_2023-08-08.png (875752e1cf5e...), FinalLightCurve\_TrES-5 b\_2023-08-08.csv (785b287dade1...)

## Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 20:27Z.